PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 4.1 (Single-Signal Drift Detectors in IDS)

Report prepared 14 August 2026

1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 4.1 (Single-Signal Drift Detectors in IDS) of the review, comprising all 16 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to this subsection's specific review question, namely whether a single, low-overhead scalar or fused signal is sufficient to detect drift and support a typed response in machine learning-based IoT or network intrusion detection, rather than against the clarity of the problem statement in the article's own, possibly different scope. A methodologically excellent article addressing a different question, such as representation-space domain adaptation, active-learning novel-class detection, malware classification, or general time-series anomaly detection, scores low or zero on QC1 here, because that problem statement is not the one this subsection screens for.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 4.1 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)

Code	Criterion	Scoring
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 16 full-text articles assessed for Section 4.1 against the quality criteria in Section 2, 9 articles met or exceeded their applicable threshold and were retained, while 7 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 4.1 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	Drift adaptive online DDoS attack detection framework for IoT system (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
2	A Network Intrusion Detection System for Concept Drifting Network Traffic Data (2021)	2021	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
3	CDDA-MD: An efficient malicious traffic detection method based on concept drift detection and adaptation technique (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
4	Intrusion detection in the IoT data streams using concept drift localization (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
5	Concept Drift Analysis Based on Isolation Forest for Effectively Detecting Network Attack in IoT Scenarios (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
6	SFC-NIDS: a sustainable and explainable flow filtering based concept drift-driven security approach for network introspection (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
7	Class imbalance and concept drift invariant online botnet threat detection framework for heterogeneous IoT edge (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
8	Self-Supervised Adaptation Method to Concept Drift for Network Intrusion Detection (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
9	Enhanced Intrusion Detection with Data Stream Classification and Concept Drift Guided by the Incremental Learning Genetic Programming Combiner (2023)	2023	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
10	VLSA-CL: A variational latent space alignment and contrastive learning framework for robust detection of IoT botnet attacks under concept drift (2026)	2026	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
11	FADES: Adaptive Drift Estimation via Conformal Signals for Streaming Intrusion Detection (2026)	2026	0.5	0.5	0.5	0.5	**2.0**	≥2.5	**Exclude**
12	DFAID: Density-aware and feature-deviated active intrusion detection over network traffic streams (2022)	2022	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
13	Fast & Furious: On the modelling of malware detection as an evolving data stream (2023)	2023	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
14	ADDAEIL: Anomaly Detection with Drift-Aware Ensemble-Based Incremental Learning (2025)	2025	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
15	Real-Time Intrusion Detection Using Adaptive Sliding Windows for Concept Drift (2025)	2025	0.5	0.5	0.0	0.5	**1.5**	≥2.5	**Exclude**
16	Anomalies Detection Using Isolation in Concept-Drifting Data Streams (2021)	2021	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**

This retained set of 9 articles constitutes the eligible pool for Section 4.1 (Single-Signal Drift Detectors in IDS) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

Prepared by
Name: RAZIMAN ZAKARIA